



Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India

(Autonomous Institute Affiliated to University of Mumbai)

Master of Computer Application Department

SYNOPTIC

End Semester Examination: 25th April -2018

Class: FYMCA Semester: II Course Code: MCA21 Branch: MCA

Name of the Course: Operating System

Q1a) An Operating System provides services to both the users and to the programs.

- It provides programs an environment to execute.
- It provides users the services to execute the programs in a convenient manner.

Following are a few common services provided by an operating system – (list down services=01marks)

Explain any 4 services in detail (04marks)

- Program execution
- I/O operations
- File System manipulation
- Communication
- Error Detection
- Resource Allocation
- Protection

b) Types of Operating Systems

Following are some of the most widely used types of Operating system.(list=01 marks)

Explain any 4 in detail (04 marks)

1. Simple Batch System
2. Multiprogramming Batch System
3. Multiprocessor System
4. Desktop System
5. Distributed Operating System
6. Network Operating System
7. Clustered System
8. Real-time Operating System
9. Handheld System

c) Demand paging: concept (02 marks)

Diagram=01 mark

Example=02 marks

d) Paging concept=01 mark

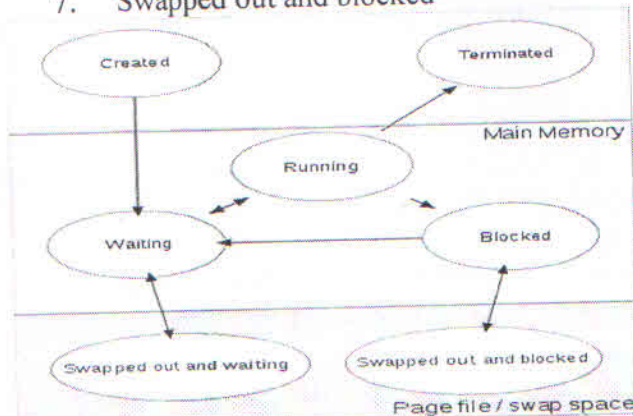
Diagram of address translation=01 marks

Explanation=03 marks

Q2b) Kernel level thread v/s User level threads (Mention any 5 points-01 mark for each)

Q2 a) In a multitasking computer system, processes may occupy a variety of states. They are as follows (list=01 mark)

1. Created
2. Ready
3. Running
4. Blocked
5. Terminated
6. Swapped out and waiting
7. Swapped out and blocked



Diagram=01 marks

States explanation=03 marks

b) Define Scheduler=01 mark

List Types of schedulers=01 mark

Explanation of working of each=03 marks

Q2c)2. Multilevel scheduling characteristics=01 mark

Multilevel feedback scheduling characteristics=01 mark

Example of each=02 marks

Diagram=01 mark

Q3 a)

From the given problem statement, it is evident that readers have higher priority than writer. If a writer wants to write to the resource, it must wait until there are no readers currently accessing that resource. (Understanding of problem statement=01 mark)

Here, we use one **mutex** and a **semaphore**. An integer variable `read_count` is used to maintain the number of readers currently accessing the resource. The variable `read_count` is initialized to 0. A value of 1 is given initially to `m` and `w`. (semaphore Initialization and purpose of each semaphore and variable =02 marks)

Instead of having the process to acquire lock on the shared resource, we use the `mutex` to make the process to acquire and release lock whenever it is updating the `read_count` variable.

The code for the **writer** process looks like this:

```

while(TRUE)
{

```

```
wait(w);
/* perform the write operation */
signal(w);
} (02 marks writers code with explanation)
```

And, the code for the **reader** process looks like this:

```
while(TRUE)
{
//acquire lock
wait(m);
read_count++;
if(read_count==1)
wait(w);
```

```
//release lock
signal(m);
```

```
/* perform the reading operation */
// acquire lock
wait(m);
read_count--;
if(read_count==0)
signal(w);
// release lock
signal(m);
} (Readers code with explanation=05 marks)
```

b) Semaphore definition=01 mark

1. Critical section problem= 03 marks (define problem=01 mark, semaphore code=01 mark, working=01 mark)
2. Define problem=01 mark, working of semaphore with example=02 marks
3. Semaphore initialization for example=01 mark, working with example=02 marks

Q3b)2. Mutual exclusion=01 mark

Hold and wait=01 mark

No-preemption=01 mark

Circular wait=01 mark

Q4b) Compare with at least 5 points(basic, fragmentation, address, size, table:01 mark for each point)

Q5a) Contiguous allocation concept=01 mark, diagram=01 mark

Linked list allocation concept=01 mark, diagram=01 mark

Indexed allocation concept=01 mark, diagram=01 mark

Q5b)1.

Disk size=32M= $2^{20} \times 32$

Block size=2K($2^{10} \times 2$)

No of blocks=disk size/size of each block= $(32 \times 2^{20}) / 2 \times 2^{10} = 16K$ (02marks)

2. FCB=purpose=01mark, diagram=01mark

Q5c)1.

Define Protection=01 mark

Access matrix concept=01 mark

Diagram=01 mark

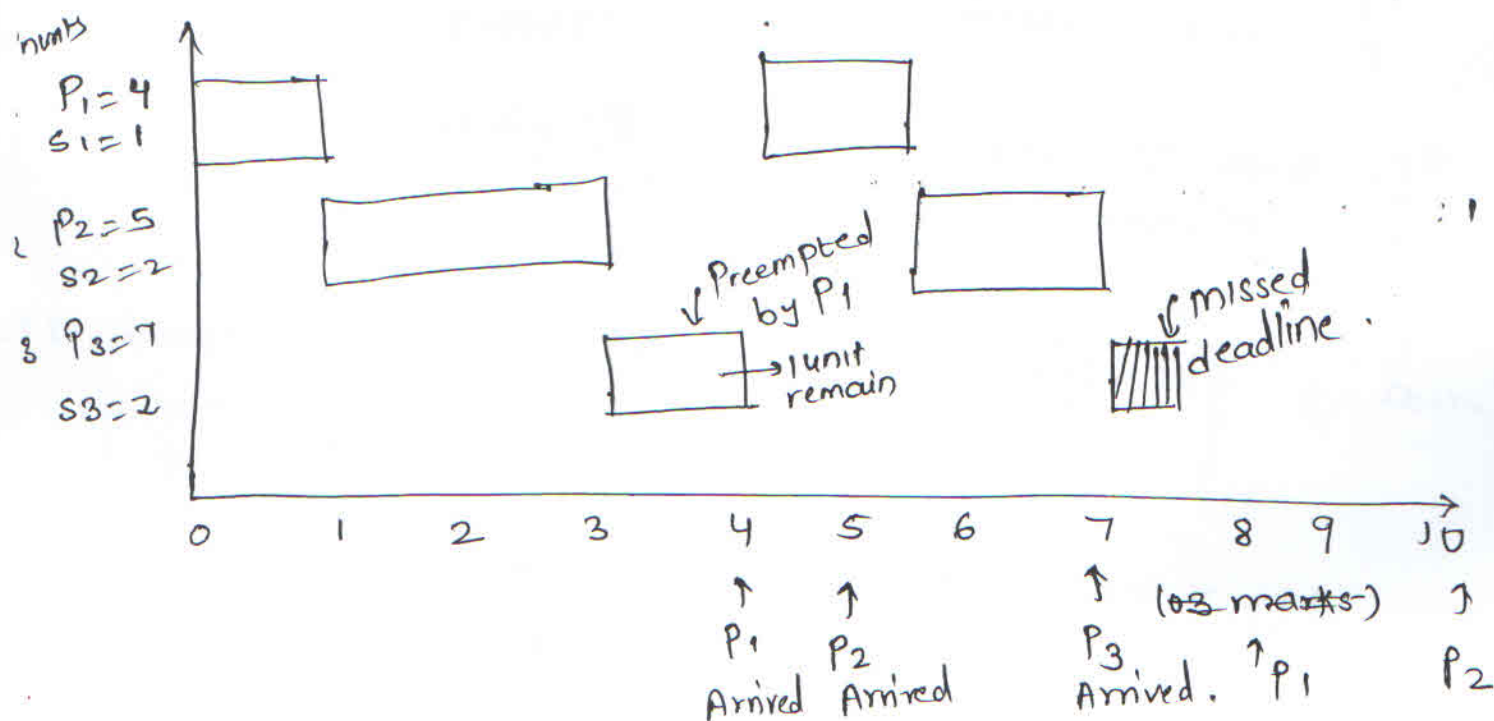
Explanation with example=03 marks

Q5 c)2.

Explanation of each RAID Level with diagram=05 mark

①

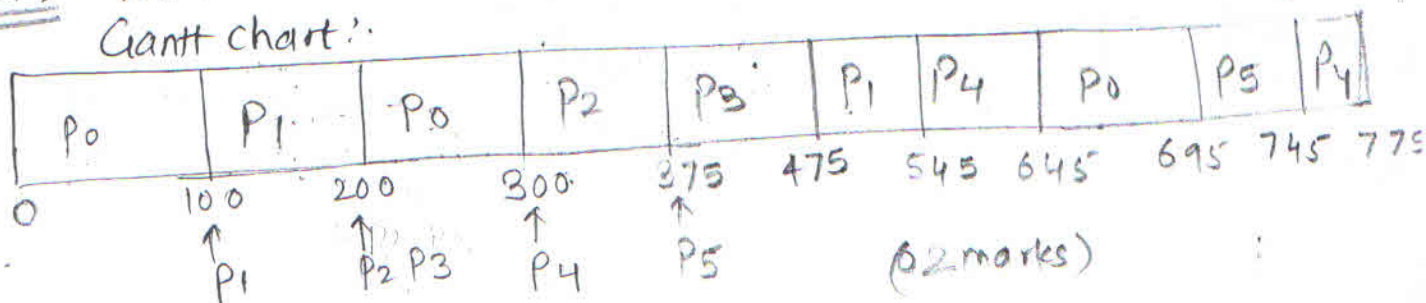
Process No	Service Time	Time Period	Priority
P ₁	1	4	1
P ₂	2	5	2
P ₃	2	7	3



- 1) Draw Chart with Suitable Notations (01 marks)
- 2) Utilization bound formula with explanation (01 marks)
- 3) Calculation of CPU utilization = 01 mark
- 4) Calculation of Utilization bound = 01 mark
- 5) Conclusion [schedulable or Not] = 01 marks using formula.

Process	AT	BT	CT	CT-AT	TAT-BT
P ₀	0	250	695	695	445
P ₁	50	170	545	495	325
P ₂	130	75	375	245	170
P ₃	190	100	475	285	185
P ₄	210	130	775	565	435
P ₅	350	50	745	395	315

1. RR with TQ = 100. Queue P₀ P₁ P₀ P₂ P₃ P₁ P₄ P₀ P₅ P₄
Gantt chart:



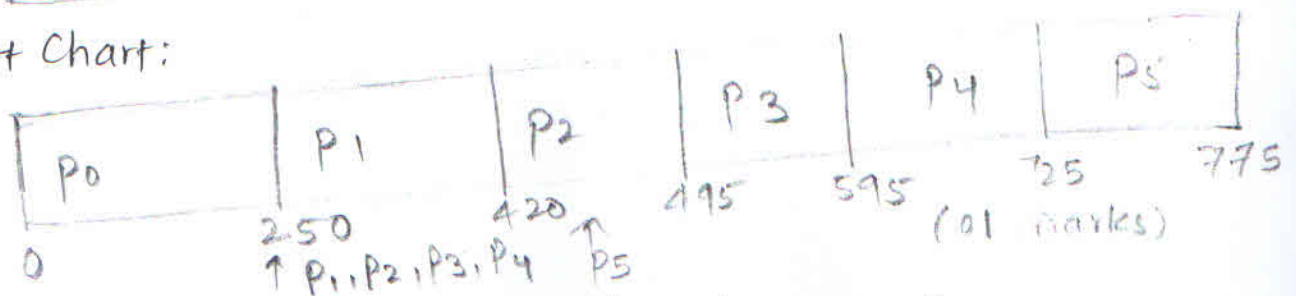
Avg TAT = 446.66 (01 mark)

Avg WT = 317.5 (01 mark)

2. FCFS:

Process	AT	BT	CT	TAT	WT
P ₀	0	250	250	250	0
P ₁	50	170	420	370	200
P ₂	130	75	495	365	290
P ₃	190	100	595	405	305
P ₄	210	130	725	515	385
P ₅	350	50	775	425	375

Gantt Chart:



Avg TAT = 388.33 (01 mark)

Avg WT = 259.16 (01 mark)

Process	AT	BT	Queue NO	CT	TAT CT-AT	WT TAT-BT
P ₁	0	4	1	6	6	2
P ₂	0	3	1	7	7	4
P ₃	0	8	2	20	20	12
P ₄	10	5	1	15	5	0

Priority of Queue 1 is greater than Queue 2

Queue 1 uses RR with TQ = 2

Queue 2 uses FCFS.

Grant chart

P ₁	P ₂	P ₁	P ₂	P ₃	P ₄	P ₃	(03 marks)
0	2	4	6	7	10	15	20

waiting Time for P₃ = 12 (02 marks).

	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P ₀	0	1	1	0	0	2	1	0	1	5	2	0
P ₁	1	2	3	1	1	6	5	2				
P ₂	1	3	6	5	2	3	6	6				
P ₃	0	6	3	2	0	6	5	2				
P ₄	0	0	1	4	0	6	5	6				

Need = Max - Allocation

(02 marks)

	A	B	C	D
P ₀	0	1	0	0
P ₁	0	4	2	1
P ₂	1	0	0	1
P ₃	0	0	2	0
P ₄	0	6	4	2

Safety Algorithm:

if (need < available)

Execute Process, new available = Currently Available + Process's Allocation.

else
Do not execute

1) need of $P_0 = (0, 1, 0, 0) < \text{Available } (1, 5, 2, 0)$

So execute P_0

$$\begin{aligned}\text{new Available} &= (1, 5, 2, 0) + (0, 1, 1, 0) \\ &= (1, 6, 3, 0)\end{aligned}$$

2) need of $P_1 = (0, 4, 2, 1)$ which is not less than $(1, 6, 3, 0)$

So Don't execute P_1

3) need of $P_2 = (1, 0, 0, 1) \nless \text{Available } (1, 6, 3, 0)$
 $\therefore$ so Don't execute P_2

4) need of $P_3 = (0, 0, 2, 0) < \text{Available } (1, 6, 3, 0)$
 $\therefore$ execute $P_3 = < P_0, P_3 >$

$$\begin{aligned}\therefore \text{new Available} &= (1, 6, 3, 0) + (0, 0, 2, 0) \\ &= (1, 6, 5, 0) = (1, 12, 6, 2)\end{aligned}$$

5) need of $P_4 = (0, 6, 4, 2) < (1, 12, 6, 2)$

So execute $P_4 = < P_0, P_3, P_4 >$

$$\begin{aligned}\therefore \text{new Available} &= (1, 12, 6, 2) + (0, 6, 4, 2) \\ &= (1, 12, 7, 6)\end{aligned}$$

6) need of $P_1 = (0, 4, 2, 1) < (1, 12, 7, 6)$
 $< P_0, P_3, P_4, P_1 >$
 $\therefore$ execute P_1

$$\begin{aligned}\text{new available} &= (1, 12, 7, 6) + (0, 4, 2, 1) \\ &= (2, 14, 10, 7)\end{aligned}$$

7) need of $P_2 = (1, 0, 0, 1) < (2, 14, 10, 7) \therefore$ execute P_2
Available =

$$\text{new Available} = (2, 14, 10, 7) + (1, 3, 6, 5) \\ = (3, 17, 16, 12)$$

④

⇒ System is in safe state

∴ safe sequence = $\langle P_0, P_3, P_4, P_1, P_2 \rangle$ (03)

Total no of resources = $(\overset{A}{3}, \overset{B}{17}, \overset{C}{16}, \overset{D}{12})$ (01)

Q.

~~current scenario of memory~~

SJF (Preemptive)

~~$P_0, P_1, P_2, P_3, P_4, P_5$~~

P_0	P_1	P_2	P_3	P_4	P_5	P_4	P_0
0	50	130	205	295	395	445	575
							775

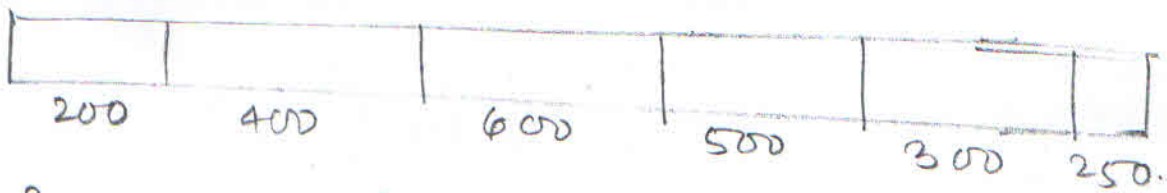
(01 marks)

	AT	BT	CT	CT-AT	TAT	WT
P_0	0	250 200	775	775	775	525
P_1	50	170 90	295	245	245	75
P_2	130	73	205	75	75	0
P_3	190	100	395	205	205	105
P_4	210	130	575	365	365	235
P_5	250	50	445	95	95	45

Avg TAT = 293.33 (01 mark)

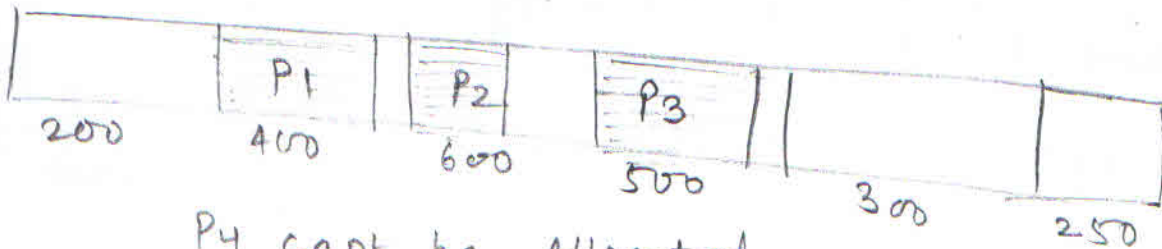
Avg WT = 164.16 (01 mark)

↳ Basic Idea - (01 mark)



$P_1 = 357K$, $P_2 = 210K$, $P_3 = 468K$, $P_4 = 491K$

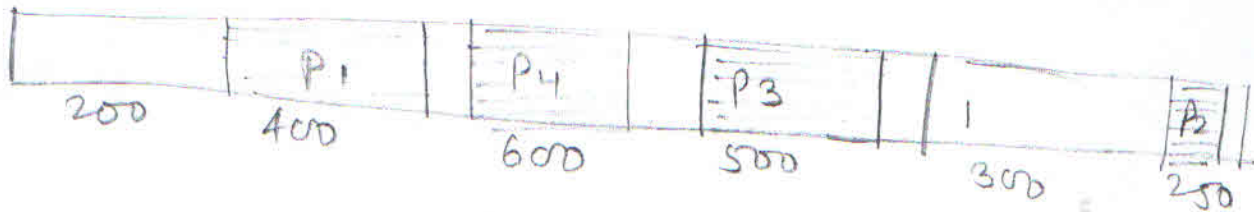
First fit:



P_4 cant be Allocated.

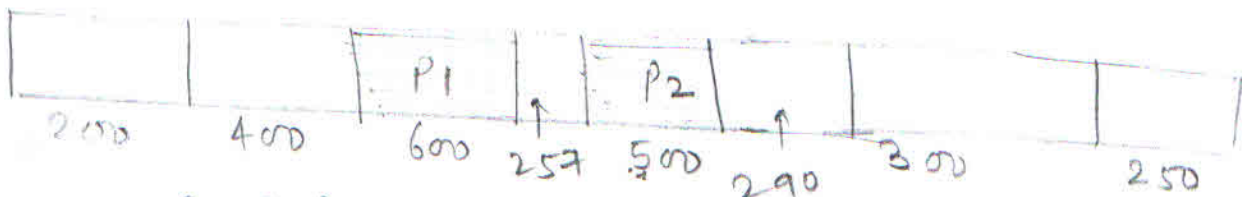
(01 mark)

Best fit



worst fit

(01 mark)



P_3 & P_4 cant be allocated.

(01 mark)

Total Space wasted due to Internal fragmentation

In worst fit = $257 + 290 = 547K$ (01 mark)

∴ In Best fit : 01 marks.

a, b, c, d, c, a, d, b, e, b, a, b, c, d, No of Frames = 4

1) FIFO

'Page fault' - 01 mark.
Definition

Frame	a	b	c	d	c	a	d	b	e	b	a	b	c	d
0	a	a	a	a	a	a	a	a	e	e	e	e	e	d
1		b	b	b	b	b	b	b	b	b	a	a	a	a
2			c	c	c	c	c	c	c	c	c	b	b	b
3				d	d	d	d	d	d	d	d	d	c	c

PH * * * *

No of Page hits = 5

No of Page faults = No of Pages - No of page hits
= 14 - 5 = 9 (03)

2. LRU

Frame	a	b	c	d	c	a	d	b	e	b	a	b	c	d
0	a	a	a	a	a	a	a	a	a	a	a	a	a	a
1		b	b	b	b	b	b	b	b	b	b	b	b	b
2			c	c	c	c	c	c	e	e	e	e	e	d
3				d	d	d	d	d	d	d	d	d	c	c

No of page hits = 7

No of page faults = 14 - 7 = 7 (03)

3. Optimal

Frame	a	b	c	d	c	a	d	b	e	b	a	b	c	d
0	a	a	a	a	a	a	a	a	a	a	a	a	a	a
1		b	b	b	b	b	b	b	b	b	b	b	b	b
2			c	c	c	c	c	c	c	c	c	c	c	c
3				d	d	d	d	d	e	e	e	e	e	e

No of page hits = 8

∴ No of page faults = 6 (03)

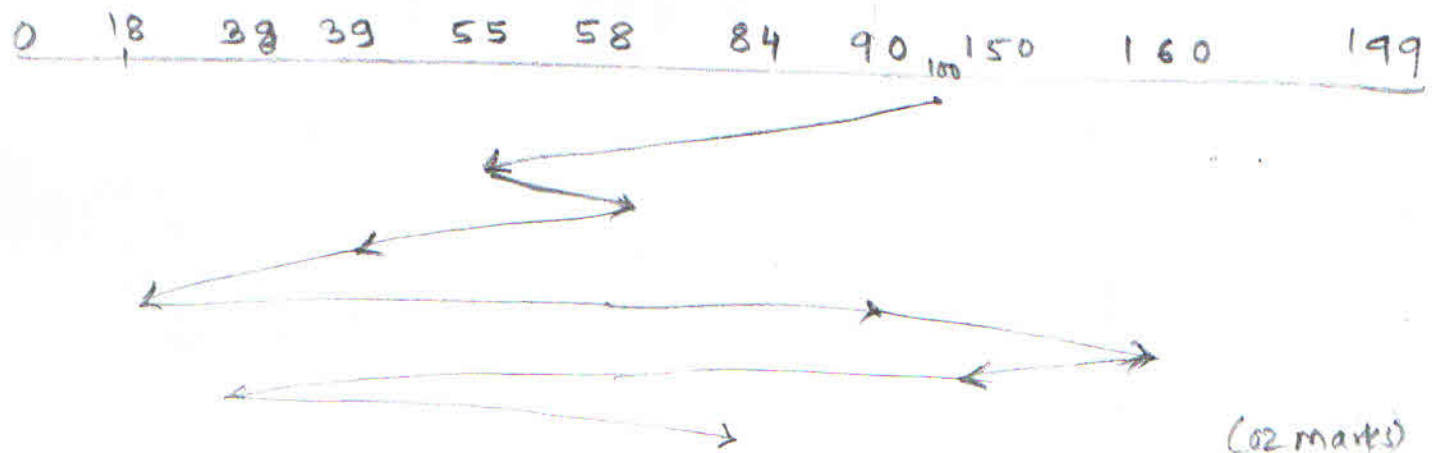
Request Queue:

55, 58, 39, 18, 90, 160, 150, 38, 84

Current head position = 100

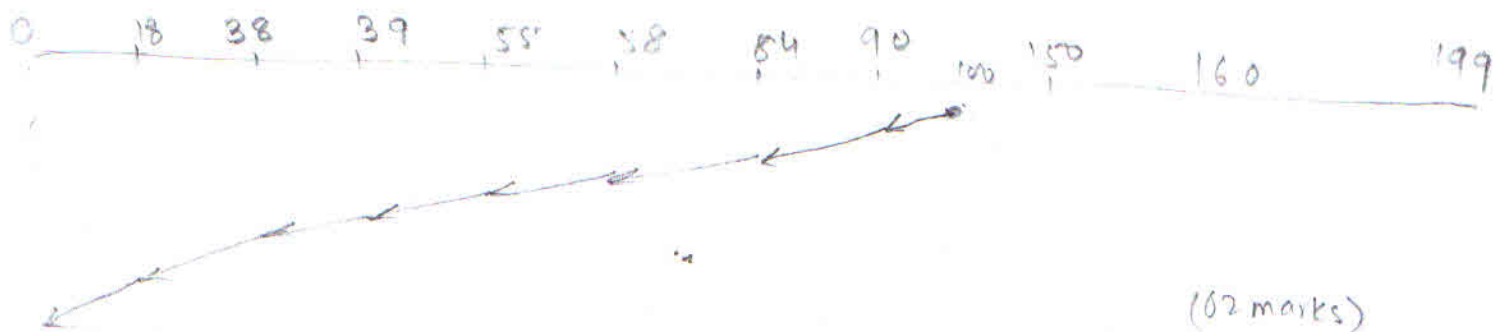
Previous -11 -11 -11 = 130

1) FCFS



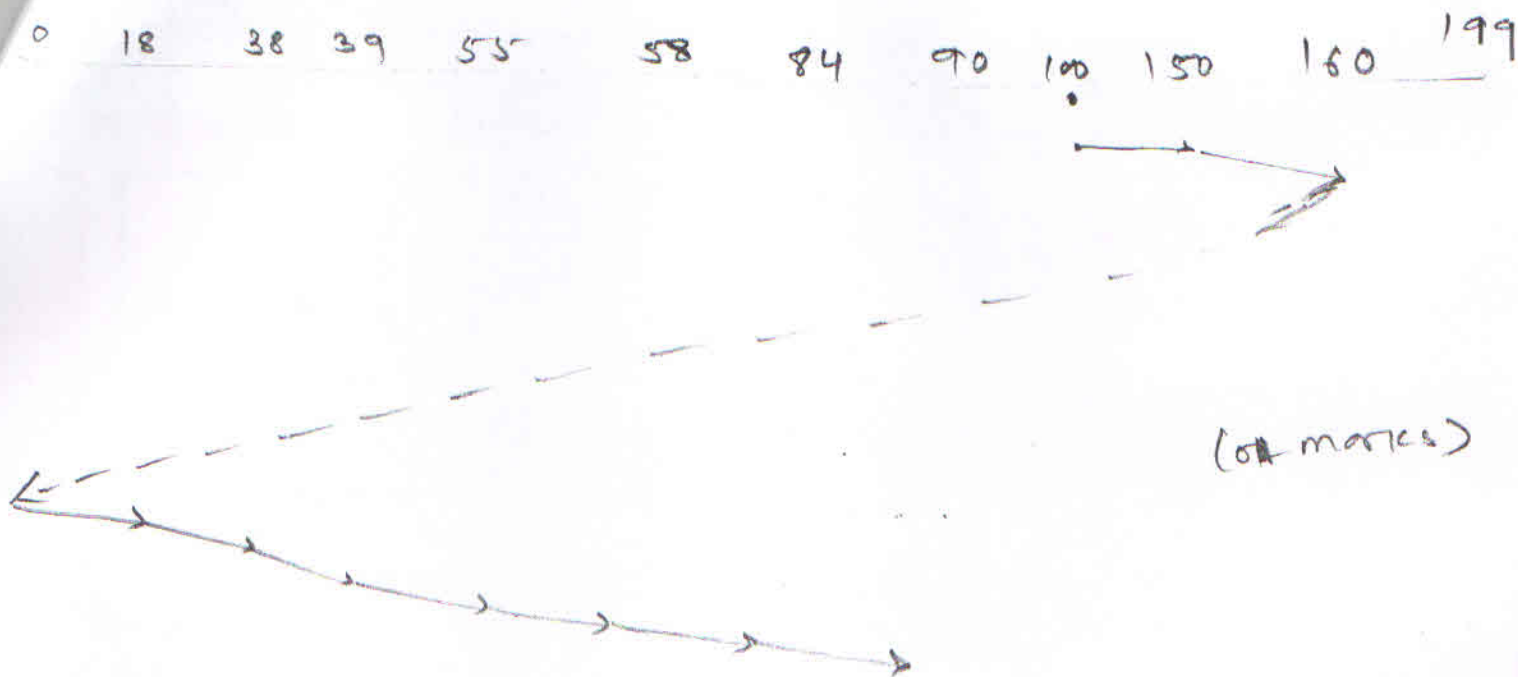
Total head movement = $(100 - 55) + (58 - 55) + (58 - 39) + (39 - 18) + (90 - 18) + (160 - 90) + (160 - 150) + (150 - 38) + (84 - 38)$
 $= 45 + 3 + 19 + 21 + 72 + 70 + 10 + 112 + 46 = 398$ (01 mark)

2) SCAN



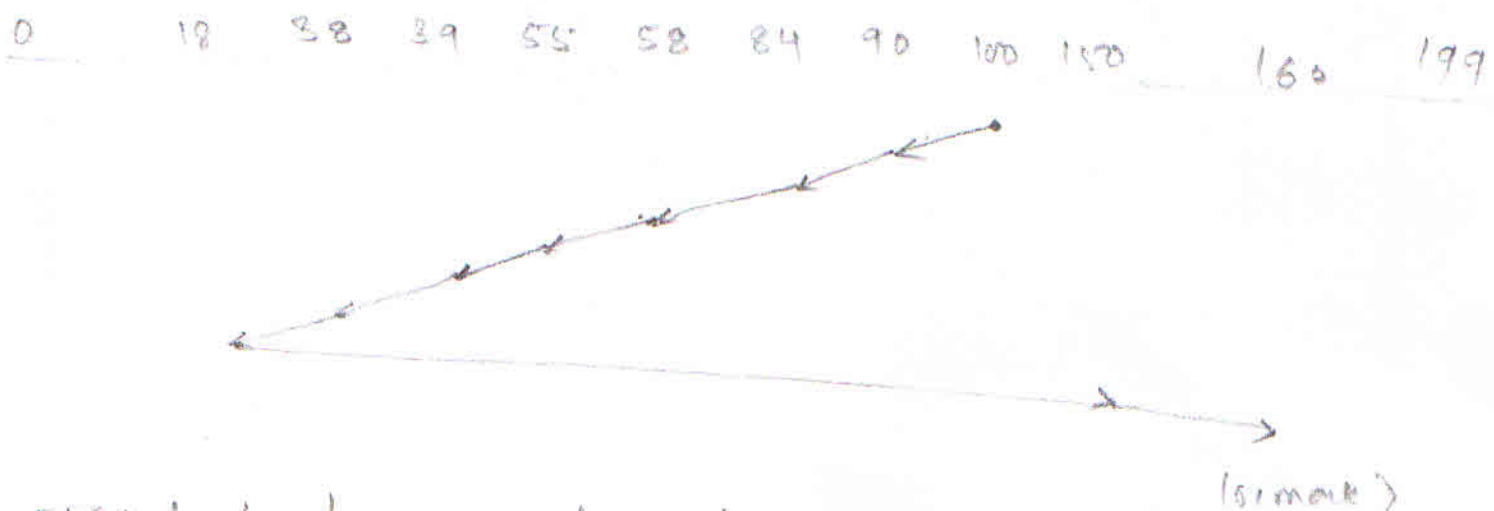
Total head movement = $(100 - 90) + (90 - 84) + (84 - 58) + (58 - 55) + (55 - 39) + (39 - 38) + (38 - 18) + (18 - 0) + (160 - 0)$
 $= 10 + 6 + 26 + 3 + 16 + 1 + 20 + 18 + 160$
 $= 260$ (01 mark)

C-SCAN



Total head movement = $(150 - 100) + (160 - 150) + (18 - 0) +$
 $(38 - 18) + (39 - 38) + (55 - 39) + (58 - 55)$
 $+ (84 - 58) + (90 - 84)$
 $= 50 + 10 + 18 + 20 + 1 + 16 + 3 + 26 + 6 = 150$
 (01 mark)

LOOK.



Total head movement = $(100 - 90) + (90 - 84) + (84 - 58) +$
 $(58 - 55) + (55 - 39) + (39 - 38) + (38 - 18) +$
 $(150 - 18) + (160 - 150)$
 $= 10 + 6 + 26 + 3 + 16 + 1 + 20 + 132 + 10 = 224$ (01 mark)